

Verifying Identities — Simplify Each Side (6.2)

Name _____

Circuit

Begin with the problem in the top left. Answer the question. To advance in the circuit, find your answer and write “2” in the blank. Solve that problem, find your answer and write “3” in the blank and so on until you have solved every problem and returned to the beginning.

ANSWER: $\tan \theta$ # _____ Simplify $\frac{1 - \cos^2 \theta}{\sin \theta}$.ANSWER: $\sec \theta$ # _____ Simplify $\sin \theta \cot \theta$.ANSWER: $\sin^2 \theta$ # _____ Simplify $\tan \theta + \cot \theta$.ANSWER: $2 \csc^2 \theta$ # _____ Simplify $(1 - \cos^2 \theta) \csc^2 \theta$.

ANSWER: $\sin \theta$

_____ Simplify $\frac{\sec^2 \theta - 1}{\sec^2 \theta}$.

ANSWER: $\cot^2 \theta$

_____ Simplify $\frac{1}{1 + \cos \theta} + \frac{1}{1 - \cos \theta}$.

ANSWER: $\cos^2 \theta$

_____ Simplify $\frac{\cos \theta}{1 - \sin^2 \theta}$.

ANSWER: 1

_____ Simplify $\frac{\sec \theta}{\csc \theta}$.

ANSWER: $\sec \theta \csc \theta$

_____ Simplify $(1 + \sin \theta)(1 - \sin \theta)$.

ANSWER: $\cos \theta$

_____ Simplify $\csc^2 \theta - 1$.